

ZKPROOF8 / ROME / DAY 2 / 16:30

Confidential Transfers for Global Payments

How Solana enables private, auditable treasury and B2B payments.

ILAN GITTER · SOLANA FOUNDATION

INTRO

Ilan Gitter

Head of Ecosystem Engineering, Solana Foundation

ZK on Solana, at a glance

NATIVE PRIMITIVES

curve25519 + MSM Edwards / Ristretto group ops, multi-scalar mul TWISTED ELGAMAL, PEDERSEN, SIGMA PROTOCOLS	ZK ElGamal Proof Program 13 sigma protocol shapes verified natively TOKEN-2022 CONFIDENTIAL BALANCES	alt_bn128 (BN254) G1 add / mul + pairing check + compression GROTH16, KZG, PLONK ON-CHAIN
Poseidon + sha256/keccak/blake3 ZK-friendly hash plus the standard set STARK / HALO2 MERKLE PATHS, EVM COMPAT	Ed25519 / Secp256k1 sigverify Native batch signature verification programs WALLET SIGS, EVM SIGNATURE COMPAT	BLS12-381 (pending mainnet) G1, G2, pairing — landing with Alpenglow / v4.0 BLS AGGREGATION, 128-BIT SNARKS

PROJECTS SHIPPING

THIS TALK Confidential Transfers	LIGHT PROTOCOL ZK Compression	NOIR ON SOLANA Sunspot	ZKVM VERIFIER RISC Zero
ENCRYPTED DEX Encipher	SHIELDED TRANSFERS Privacy Cash	PRIVATE TRADING Yona	ENCRYPTED DEX Darklake

PROBLEM

“Can anyone see our business flows onchain?”

Today's honest answer on public chains: yes.

Public amounts leak business strategy

Cross-border settlement

Corridor volumes, FX positioning, and per-transfer amounts visible to anyone watching the ledger.

Stablecoin mint and burn

Issuer working capital and processor settlement flows readable in real time.

Payroll and vendor payments

Salary bands, supplier terms, and contracted discounts exposed in transfer history.

Treasury rebalancing

Liquidity provisioning between custodians and regions reveals strategy before execution.

Interbank settlement

Counterparty pairs and amounts visible to competitors and chain analysts.

COUNTERPARTY VISIBILITY

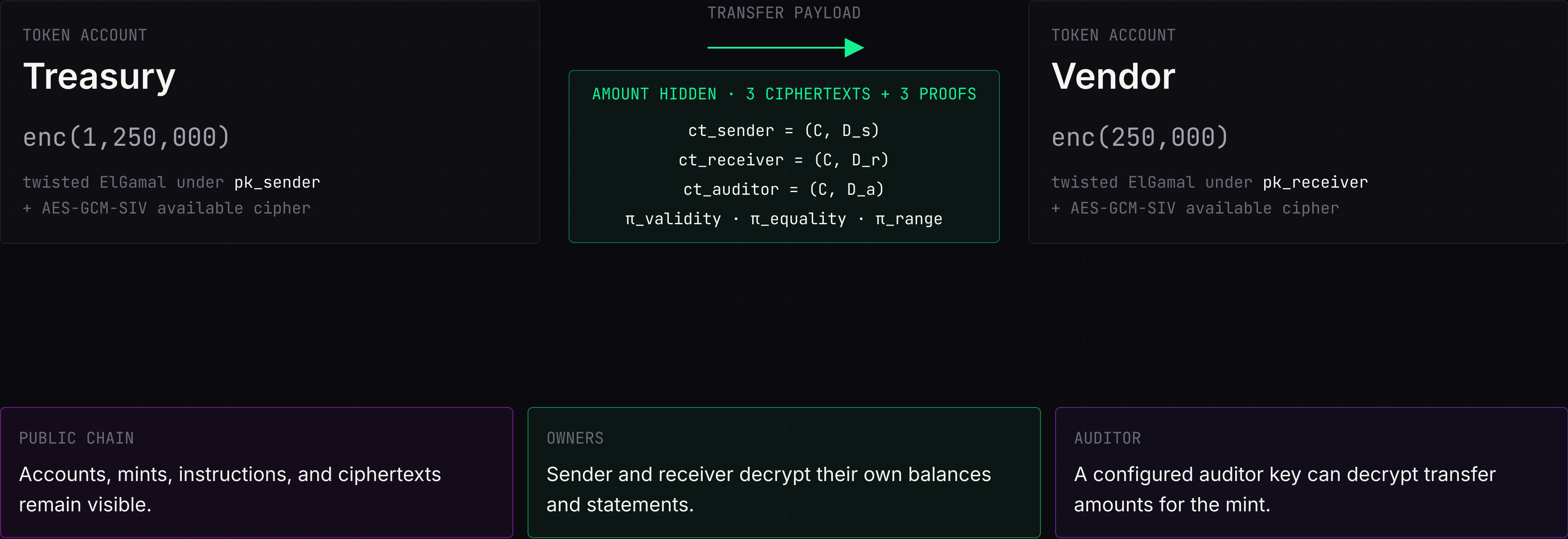
ACH	AMOUNTS HIDDEN
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SWIFT	AMOUNTS HIDDEN
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Card networks	AMOUNTS HIDDEN
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Public chains today	AMOUNTS PUBLIC
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Confidential transfers at a glance



Protocol overview

- 01 CONFIGURE MINT
- 02 OPT-IN
- 03 DEPOSIT
- 04 APPLY PENDING
- 05 TRANSFER
- 06 WITHDRAW

CONFIGURE MINT

Issuer enables confidential transfers

Mint is created with the ConfidentialTransferMint extension. Issuer optionally sets an auditor ElGamal pubkey and a privacy level (disabled / opt-in / whitelisted / required).

PRIMITIVES

twisted ElGamal pk_auditor

policy flag

PUBLIC CHAIN

extension data, auditor pubkey

OWNER

issuer policy is fixed

AUDITOR

registered audit target

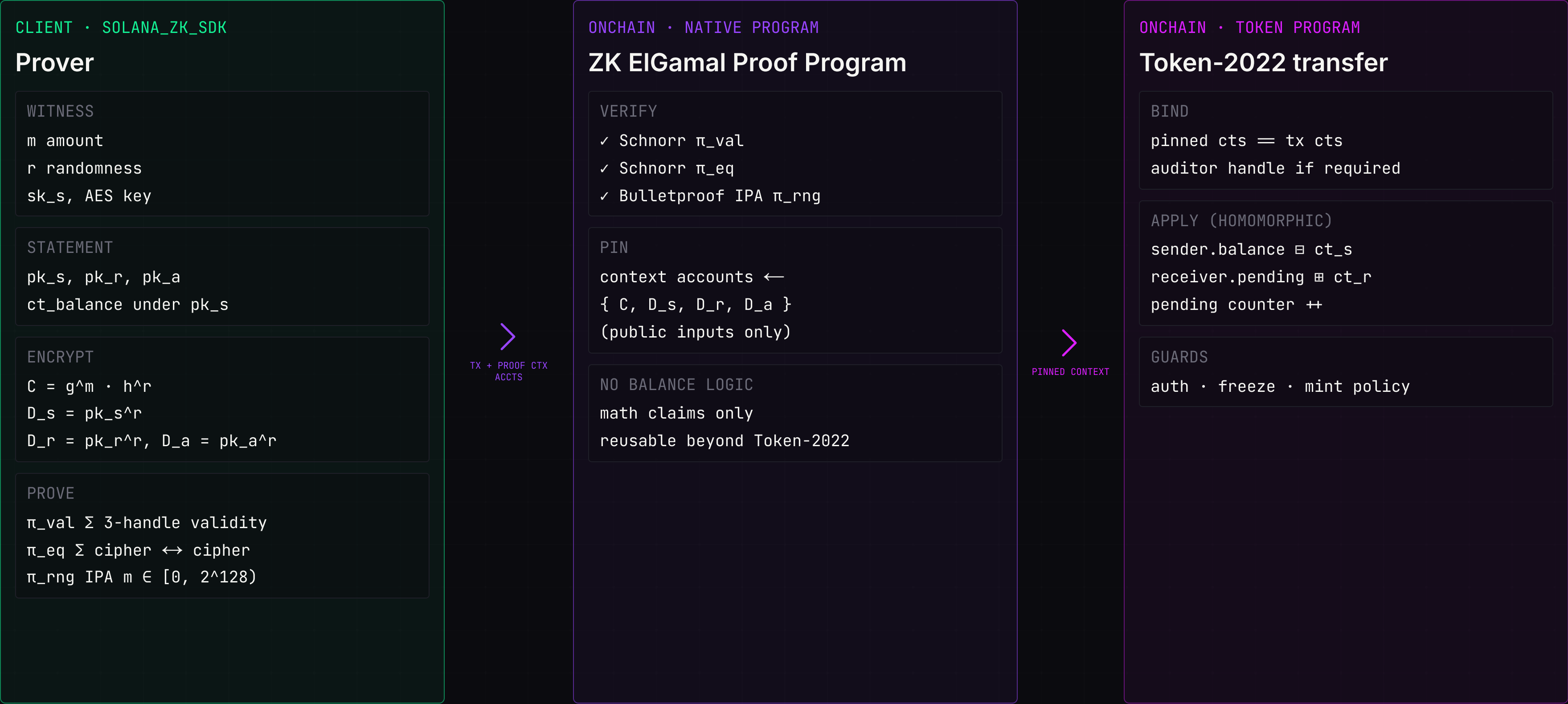
Protocol vs paper

PRIMITIVE	PAPER	USED FOR	SDK
Twisted ElGamal encryption	Anza twisted-elgamal notes; Chen, Ma, Li	Encrypts every balance and transfer amount	<code>solana_zk_sdk::encryption::elgamal</code>
Pedersen commitments	Pedersen 1991	Hides amounts inside ElGamal ciphertexts and proof witnesses	<code>embedded in every ElGamal ciphertext</code>
Sigma protocols (Fiat-Shamir)	Schnorr 1989; Maurer 2009	Pubkey validity, zero-ciphertext, equality	<code>PubkeyValidity, ZeroCiphertext, CiphertextCiphertextEquality</code>
Grouped multi-recipient validity	Sigma extension over grouped ciphertexts	Sender + receiver + auditor encryptions match	<code>GroupedCiphertext3HandlesValidity</code>
Bulletproof range proofs	Bünz, Bootle, Boneh et al. 2018	Hidden amount fits in $[0, 2^{64})$ — no negative spend	<code>BatchedRangeProofU64 / U128</code>

NOT IN THE PAPERS

- AES-GCM-SIV second ciphertext on every available balance — owner reads without solving DLP
 - ZK ElGamal Proof Program as a separate Solana program, CPI'd from Token-2022
- Proof context state accounts — large proofs pass by reference, transfer fits in one tx
 - 16-bit lo / 16-bit hi ElGamal split on pending balance for tractable u32 decryption

Proof flow



Adoption hurdles

Ecosystem adoption

Issuers, wallets, exchanges, and indexers all have to learn a new account shape. Standards work is in progress; coverage is not yet uniform.

TOKEN EXTENSIONS SUPPORT ACROSS THE STACK IS THE GATING ITEM.

Transaction sizes

Equality, ciphertext-validity, and range proofs do not fit in one transaction. They go in proof context state accounts: create, reference, close.

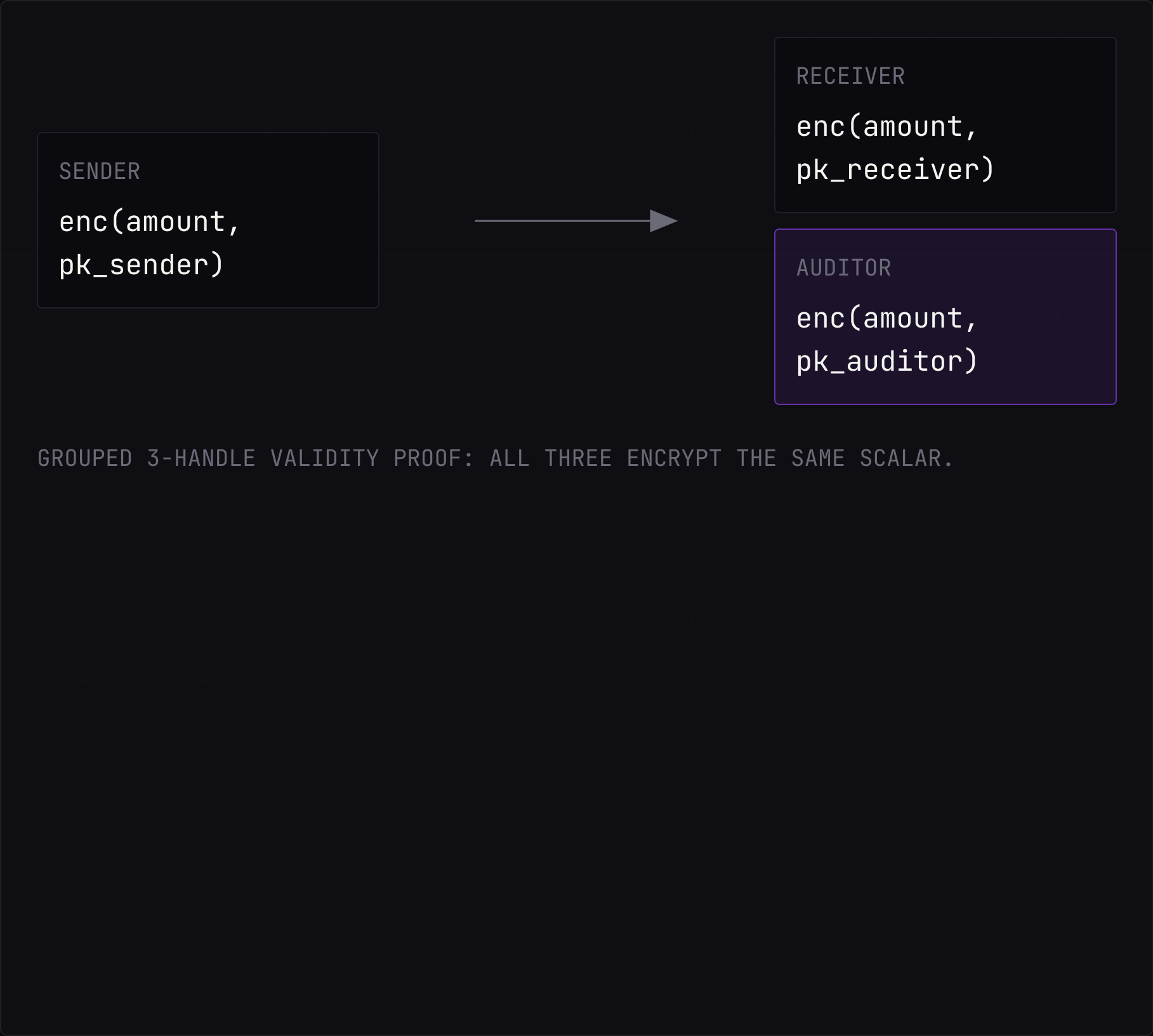
A CONFIDENTIAL TRANSFER IS 3 TRANSACTIONS ON THE WIRE TODAY.

Composability

Programs that need to read amounts (AMMs, lending, oracles) cannot read encrypted balances directly. Most flows reveal at the boundary, then re-shield.

ENCRYPTED-INPUT COMPOSABILITY IS AN OPEN PROBLEM.

Auditor key



WHAT IT GIVES

- Mint-level decryption of every transfer amount
- Issuer-side suspicious activity review
- Compliance story that full anonymity cannot offer

INSTITUTIONS STILL NEED

- Internal selective disclosure protocol (per-counterparty, per-transaction)
- Regulatory clarity on auditor scope, retention, and chain of custody
- Long-term key management story (rotation, threshold control, revocation)

Demo setup

Devnet

Live cluster running.

SETUP COMPLETED

01

Configure mint and accounts

02

Opt-in to confidential transfers

03

Mint USDC to treasury account

Live transfer / selective disclosure

OFFLINE

OWNER DECRYPT Sender / Treasury	OWNER DECRYPT Receiver / Vendor	GLOBAL DECRYPT Auditor / Regulator	NO KEY Chain analyst / Public
CIPHERTEXT enc(...) DECRYPTED —	CIPHERTEXT enc(...) DECRYPTED —	CIPHERTEXT — DECRYPTED —	CIPHERTEXT enc(...) DECRYPTED -- not authorized --

START BACKEND: CARGO RUN --BIN DEMO-SERVER FAILED TO FETCH RUN R APPLY A

What payment companies can build

Treasury rebalancing

Move liquidity between issuers, regions, and custodians without broadcasting strategy.

GLOBAL BANKS, MONEY-TRANSFER NETWORKS, TREASURY OPERATORS

B2B settlement

Vendor, payroll, intercompany, and cross-border flows with public settlement and private amounts.

PAYMENT PROCESSORS, PAYROLL PLATFORMS, CROSS-BORDER NETWORKS

Stablecoin operations

Issuer and processor flows where the market should not learn working capital in real time.

STABLECOIN ISSUERS, PROCESSORS, FINTECHS

Beyond payments

SOLANA PRIMITIVE

Encrypted value plus public verification becomes a substrate, not just a payments feature.

The same ZK ElGamal Proof Program that backs confidential balances also backs trading, securities settlement, and governance: order books, auctions, DVP, regulated securities, encrypted votes.

Private order books

Commit to order state without exposing inventory or intent before execution.

EXCHANGES, MARKET MAKERS, REGULATED SECURITIES DESKS

Sealed-bid auctions

Encrypted bids with public settlement and proofs that the clearing rule was followed.

INVESTMENT FIRMS, DVP / SETTLEMENT, RWA PLATFORMS

Encrypted governance

Votes, allocations, and state transitions that reveal outcomes without exposing every input.

DAOS, INSTITUTIONAL VOTING, ALLOCATION COMMITTEES

Open problems for this room

GOALS

Solana optimizes for reduced onchain state and computation complexity. The ecosystem is prioritizing privacy and composibility with these constraints.

Private trading

Sealed orders, threshold decryption at fill, MEV-resistant matching engines.

Programmable encrypted state

selective homomorphic ops, encrypted state transitions, threshold decryption.

Aggregation primitives

Folding, IPA, sumcheck. Batched confidential updates.

Hybrid stacks

TEE, MPC, and ZK combined. Pure cryptographic privacy is often too slow for Solana's latency target.

Private composability

AMMs, lending, liquidation under encryption.

QUESTIONS

Connect

X x.com/nocircuit

GITHUB github.com/gitteri

SLIDES • [GITHUB.COM/GITTERI/ZKPROOF8-TALK](https://github.com/gitteri/zkproof8-talk)